

QUESTION: Develop a Java program to create a menu-driven “Library Management System” that demonstrates the core Object-Oriented Programming concepts: Classes and Objects, Constructors, Encapsulation, Inheritance, Method Overloading, Method Overriding, Polymorphism, Abstraction, and Interfaces.

Steps for Execution

1. Open IntelliJ IDEA / Eclipse / VS Code.
2. Create a Java project.
3. Create a package named javaprograms.
4. Create a class named LibraryManagementSystem.
5. Enter the given code and save it.
6. Run the LibraryManagementSystem class.
7. The program creates book, student, faculty, vehicle, shape, and report objects.
8. The corresponding details and operations are displayed.

Algorithm

1. Start.
2. Create a Books object and demonstrate encapsulation using getters and setters.
3. Create another Books object using a parameterized constructor.
4. Create Student1 and Faculty1 objects to demonstrate inheritance.
5. Create an Areas object and call overloaded area() methods for square, rectangle, and circle.
6. Create Cars and Bikes objects and call the overridden display() method.
7. Create Circles and Rectangles objects from the abstract Shapes class and call draw().
8. Create a Reports object implementing the Prints interface.
9. Call print() to display the library report.
10. Stop.

UML Class Diagram



```

| - bookId : int |
| - bookName:String |
| - author : String |
| - price : double |
|_____|
| + getters/setters |
| + display() |
|_____|

```

```

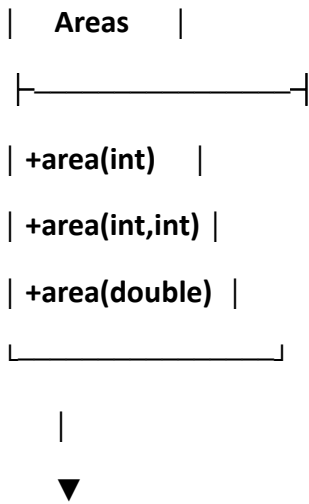
|_____|
| Person1 |
|_____|
| name : String |
| age : int |
|_____▲_____|
|
|
|_____|
| |
|_____| |_____|
| Student1 | | Faculty1 |
|_____| |_____|
| rollNo | | subject |
| +display | | +display |
|_____| |_____|

```

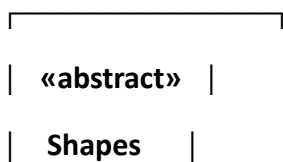
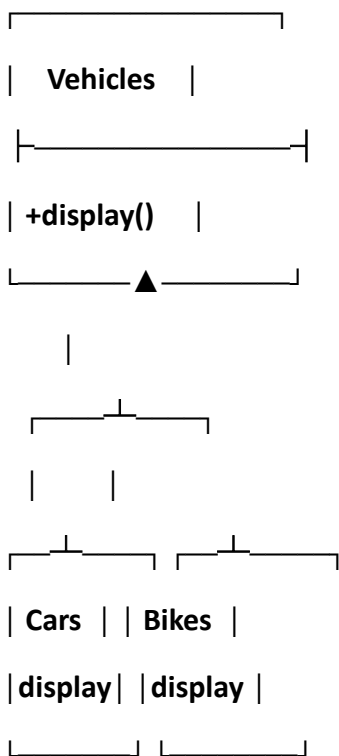
```

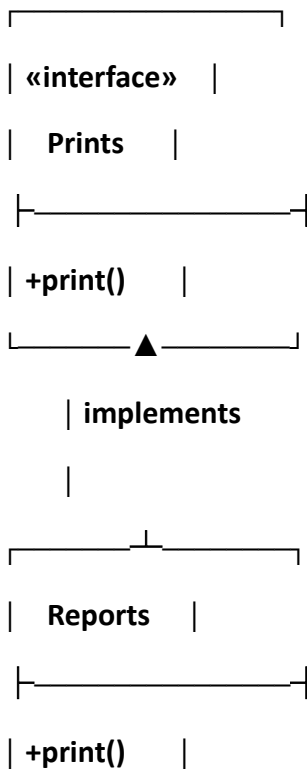
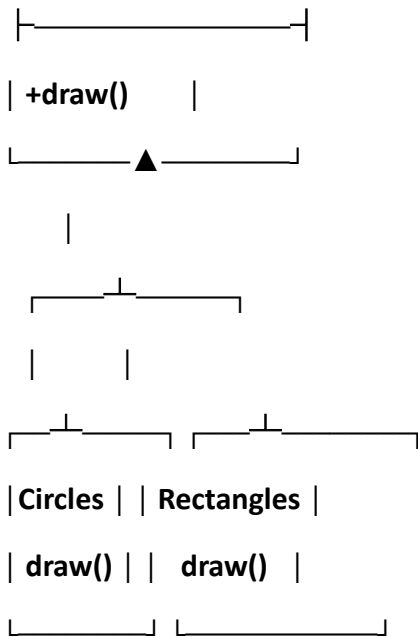
|_____|

```



Method Overloading





CODE:

```

class Books {

    private int bookId;
    private String bookName;
    private String author;
  
```

```
private double price;

public Books(int id, String name, String author, double price) {
    this.bookId = id;
    this.bookName = name;
    this.author = author;
    this.price = price;
}

//getters Setters
public void setBookId(int id) {
    bookId = id;
}

public void setBookName(String name) {
    bookName = name;
}

public void setAuthor(String a) {
    author = a;
}

public void setPrice(double p) {
    price = p;
}

// Getters
public int getBookId() {
    return bookId;
}

public String getBookName() {
    return bookName;
}

public String getAuthor() {
    return author;
}

public double getPrice() {
    return price;
}

public void display() {
    System.out.println("Book ID : " + bookId);
    System.out.println("Book Name : " + bookName);
    System.out.println("Author : " + author);
}
```

```
        System.out.println("Price : " + price);
    }
}

class Person1 {
    String name = "priya";
    int age = 22;
}

class Student1 extends Person1 {
    int rollNo = 156;

    void displayStudent() {
        System.out.println("\nStudent Details");
        System.out.println("Name : " + name);
        System.out.println("Age : " + age);
        System.out.println("Roll No : " + rollNo);
    }
}

class Faculty1 extends Person1 {
    String subject = "Java";

    void displayFaculty() {
        System.out.println("\nFaculty Details");
        System.out.println("Name : " + name);
        System.out.println("Age : " + age);
        System.out.println("Subject : " + subject);
    }
}

class Areas {

    // Method Overloading
    void area(int side) {
        System.out.println("\nArea of Square = " + (side * side));
    }

    void area(int length, int breadth) {
        System.out.println("Area of Rectangle = " + (length * breadth));
    }

    void area(double radius) {
        System.out.println("Area of Circle = " + (3.14 * radius * radius));
    }
}
```

```
// Method Overriding
```

```
class Vehicles {  
    void display() {  
        System.out.println("\nThis is a Vehicle");  
    }  
}
```

```
class Cars extends Vehicles {  
    void display() {  
        System.out.println("This is a Car");  
    }  
}
```

```
class Bikes extends Vehicles {  
    void display() {  
        System.out.println("This is a Bike");  
    }  
}
```

```
abstract class Shapes {  
    abstract void draw();  
}
```

```
class Circles extends Shapes {  
    void draw() {  
        System.out.println("\nDrawing Circle");  
    }  
}
```

```
class Rectangles extends Shapes {  
    void draw() {  
        System.out.println("Drawing Rectangle");  
    }  
}
```

```
interface Prints {  
    void print();  
}
```

```
class Reports implements Prints {  
    public void print() {  
        System.out.println("\nPrinting Library Report");  
    }  
}
```

```
public class LibraryManagementSystem {
```

```
public static void main(String[] args) {
    Books b1 = new Books(167,"Java Programming","James Gosling",599);

    b1.setBookId(167);
    b1.setBookName("Java Programming");
    b1.setAuthor("James Gosling");
    b1.setPrice(599);

    System.out.println("\nBook Details");
    b1.display();

    Books b2 = new Books(176, "Python", "Guido", 700);

    System.out.println("\nParameterized Constructor");
    b2.display();
    Student1 s = new Student1();
    s.displayStudent();

    Faculty1 f = new Faculty1();
    f.displayFaculty();

    // Method Overloading
    Areas a = new Areas();
    a.area(6);
    a.area(10, 24);
    a.area(3.7);

    // Method Overriding
    Vehicles v;

    v = new Cars();
    v.display();

    v = new Bikes();
    v.display();
    Shapes sh;

    sh = new Circles();
    sh.draw();

    sh = new Rectangles();
    sh.draw();
    Reports r = new Reports();
    r.print();
}
}
```


OUTPUT:

Book Details

Book ID : 167

Book Name : Java Programming

Author : James Gosling

Price : 599.0

Parameterized Constructor

Book ID : 176

Book Name : Python

Author : Guido

Price : 700.0

Student Details

Name : priya

Age : 22

Roll No : 156

Faculty Details

Name : priya

Age : 22

Subject : Java

Area of Square = 36

Area of Rectangle = 240

Area of Circle = 42.9866

This is a Car

This is a Bike

Drawing Circle

Drawing Rectangle

Printing Library Report